RENEWABLE ENERGY





Renewable energy is growing in importance and popularity

- The desire and necessity to avert irreversible climate damage.
- Increasing oil prices.
- The unreliability of non-renewable resources (e.g., the depletion of oil wells). considering these and other factors, governments worldwide support renewable energy with various incentives. This encourages entrepreneurs to make large-scale investments in renewable energy.

There are two main types of solar Energy technologies –

Photovoltaic (PV)

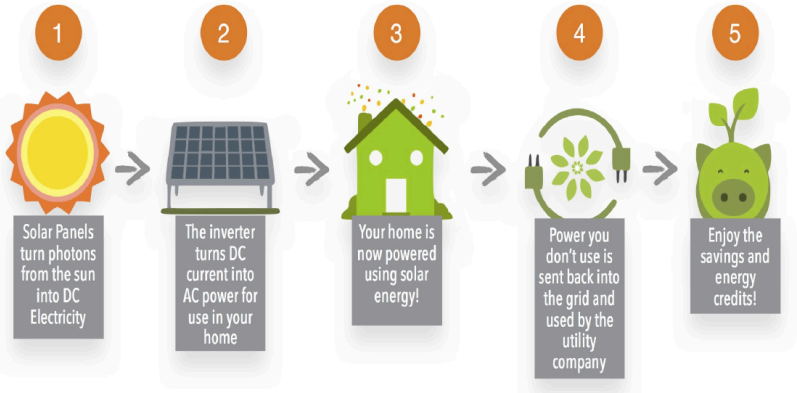
Photovoltaic solar energy is obtained by converting sunlight into electricity using a technology based on the photoelectric effect. It is a renewable, inexhaustible, and non-polluting energy that can be produced in installations ranging from small generators for self-consumption to large photovoltaic plants.

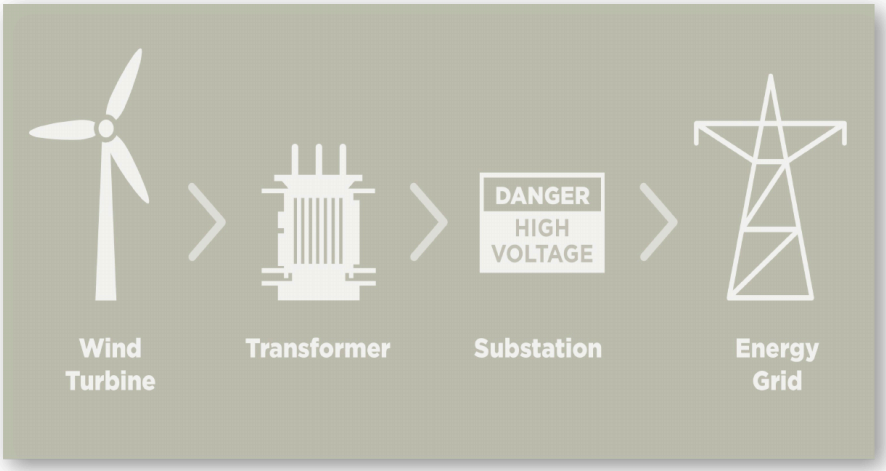


Concentrating solar-thermal power (CSP)

Concentrating solar-thermal power (CSP) technologies can be used to generate electricity by converting energy from sunlight to power a turbine. Still, the same basic technologies can also be used to deliver heat to a variety of industrial applications, like water desalination, enhanced oil recovery, food processing, chemical production, and mineral processing.





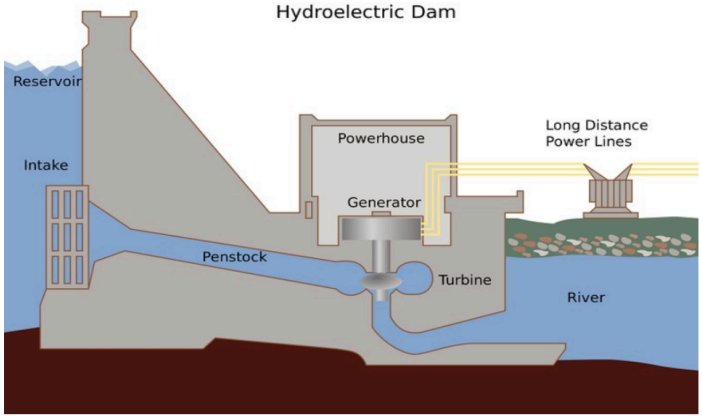


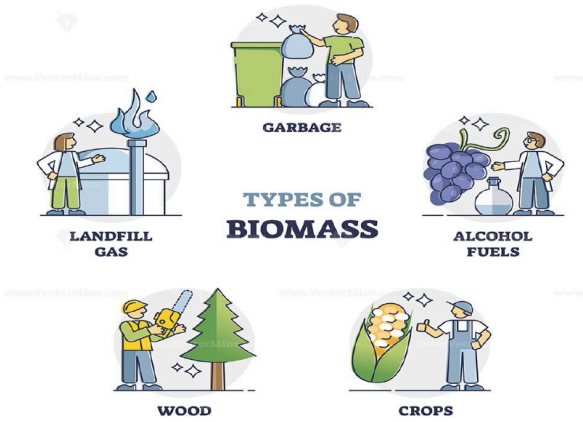
HYDRO ENERGY

Hydropower, or hydroelectric power, is a renewable energy source that uses a dam or diversion structure to alter the natural flow of a river or other body of water.



HOW DOES HYDRO ENERGY WORK?



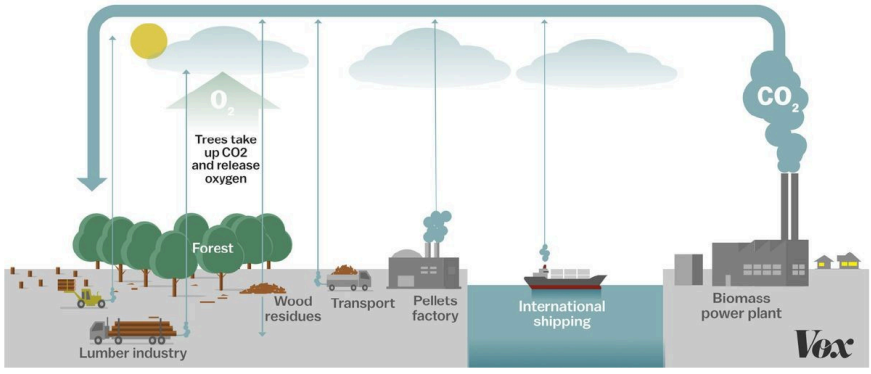


Biomass is a plant-based material used as fuel to produce heat or electricity. Examples are wood and wood residues, energy crops, agricultural residues, and waste from industry, farms, and households. Since biomass can be used as a fuel directly, some people use the words biomass and biofuel interchangeably.



Biomass is probably our oldest source of energy after the sun.

The cycle of biomass energy



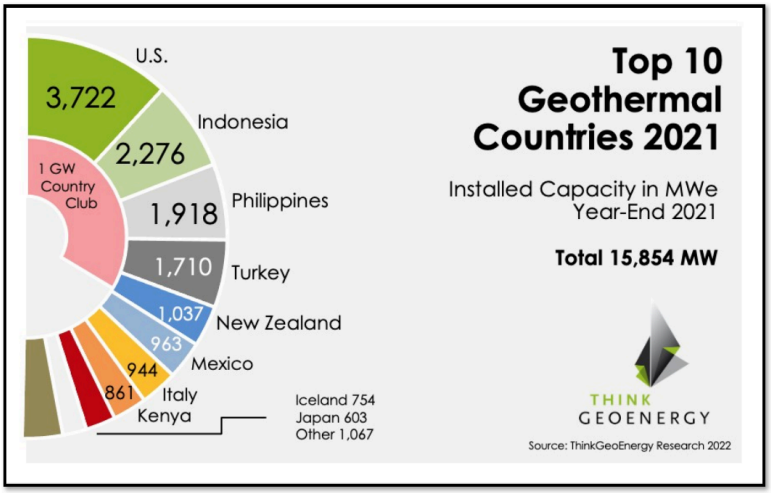
Perks of Geo Energy-

1. Baseload energy - it's always on: Geothermal power plants produce electricity consistently, running 24 hours a day, 7 days a week. The power output of a geothermal power plant is highly predictable and stable, thus facilitating energy planning with remarkable accuracy.

2. Geothermal energy can heat, cool, and generate electricity: Geothermal energy can be used in different ways depending on the resource and technology chosen—heating and cooling buildings through geothermal heat pumps, generating electricity through geothermal power plants, and heating structures through direct-use applications.

3. Over 100 years of geothermal sourced energy: The first ever geothermal plant was set up in Larderello, Italy 1904. Steam from that geothermal source was used to turn a small turbine that powered five light bulbs. Today, the U.S. leads the world in geothermal power generation, providing more than 3.7 GW to the national grid.

4. Weather Independence: A significant advantage is that geothermal energy is not dependent on weather conditions.

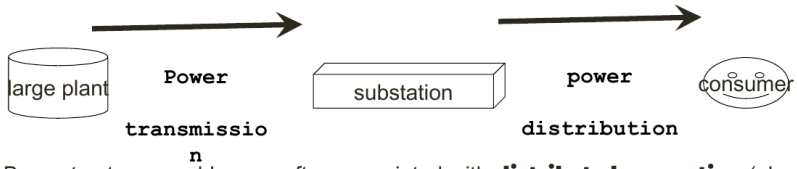


ADVANTAGES & DISADVANTAGES OF RENEWABLE ENERGY

- ❑ Easily regenerated
- ❑ Supports environment
- ❑ Easily available
- ❑ Low maintenance cost
- ❑ Weather dependency
- ❑ High installation cost
- ❑ Fluctuation problem(Solar)
- ❑ Geographic limitations

Distributed generation

Traditional energy generation is mostly done in centralized facilities, from where energy must travel a long way to reach the end consumer:



By contrast, renewables are often associated with **distributed generation** (also called **dispersed generation** or **decentralized energy**). This is producing energy in many small facilities and transporting it over short distances. Roof solar panels and wind turbines are examples of distributed energy resources (DERs).

For renewable energy to become massively used, energy systems must be adjusted to reflect the shift from centralized to dispersed generation.

What energy qualifies as renewable



Some **scientists and politicians argue** that **nuclear energy is renewable** since the resources from which it is derived (such as uranium) would not be exhausted in millions of years.



These claims, **however, have not been proven**; furthermore, nuclear energy has an extremely dangerous byproduct: **nuclear waste**. For this reason, governments don't recognize nuclear energy as renewable, and it is not eligible for state subsidies.



Fossil fuels could be regarded as biomass since they have biological origin; **however, they are neither sustainable nor green because**:

- this is organic material that has undergone millennium-long geological transformation;
- thus, the regeneration rates of fossil fuels are extremely slower than the rate at which they are consumed;
- fossil fuels emit CO_2 when burnt.